The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform

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- : The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform
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- : O-RAN, 5G, AI, xApp, RIC, SAC

1.1

AI-RAN Alliance **AI-and-RAN**(AI workload RAN workload) , O-RAN NRT-RIC **CAORA** , GPU (MIG) workload **SAC(Soft Actor-Critic)** . RAN workload 99% RAN 100% .

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2 (Motivation)

RAN network function **dedicated** . (static) .

1. **Off-peak** : RAN .
2. **AI workload** : GenAI/edge AI inference .

workload .

	RAN workload	AI workload
	latency-sensitive, (non-elastic) carrier ( )	(elastic) use case best-effort

, . **RAN** **AI workload** , **(coexistence)** .

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3 CAORA

CAORA (Converged AI-and-ORAN Architecture) O-RAN 4 .

3.1

- **Layer 1 — External Cloud**: AI/ML workload . edge AI inferencing, generative AI, robotics application , **AI Deployment API** RAN (compute cluster) .
- **Layer 2 — Programmable RAN**: O-RAN disaggregation O-RU / O-DU / O-CU.
- **Layer 3 — Controller (NRT-RIC)**: traffic prediction, resource allocation, monitoring custom xApp . **E2 interface** RAN .
- **Layer 4 — Resource Management (E2E Orchestrator)**: compute cluster(GPU, **MIG**, NIC) . SAC agent .

3.2 : Y1 interface

Y1 interface .

- **Monitoring xApp** (Layer 3) bandwidth utilization, system latency, network load RAN KPI .
- radio analytics **Y1 interface** E2E orchestrator(Layer 4) .
- E2E orchestrator **Y1 consumer** SAC .

“xApp = , E2E orchestrator(SAC) = ” , SAC agent : (i) **(predictive analysis)**, (ii) **(real-time adaptation)**, (iii) peak RAN **(priority management)**.

RAN network AI workload **multi-purpose system** .

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4.1 Task

task K, task $k \in K$:

$$k = \{\tau, r, p\}$$

- τ: task (RAN / AI)
- $r = [r_i, \ \forall i = 1, 2, \ldots, N]$: (resource type)
- $p \in [0, 1]$: (> /)

$r(k,t)$, $p(k)$.

4.2

task k t L_1 norm .

$$\Delta(k,t) = \|r(k,t)\|_1 = \sum_{i=1}^{n} r_i(k,t)$$

task .

$$\Delta_{\text{total}}(t) = \sum_{k \in K} \Delta(k,t)$$

4.3 (System Dynamics)

workload $W(t)$ task , **(contention)** .

$$W(t) = \int_0^t \left(\sum_{k \in K} P(k,t)\,\Delta(k,t) \;-\; \sum_{k \in K} P_c(k,t)\,\Delta(k,t)\,C(k,t) \right) dt$$

- $P(k,t)$: task
- $P_c(k,t) = \alpha \cdot p(k)$: task ($\alpha$, $p(k)$)
- $C(k,t)$: task k

, ( ) .

4.4

RAN AI , . .

$$r_{\text{RAN}}(t) + r_{\text{AI}}(t) \;\leq\; \mathcal{R}_{\max}$$

$\mathcal{R}_{\max}$, NVIDIA MIG ( ).

5 SAC

E2E orchestrator **SAC(Soft Actor-Critic)** . SAC entropy **-** , . .

- **State**: monitoring xApp RAN/AI (bandwidth utilization, system latency, network load).
- **Action**: RAN AI ( ).
- **Reward**: task , **(utilization)** . peak RAN .

"RAN " "AI workload best-effort " .

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6.1

- **Python** (O-RAN).
- **NVIDIA MIG(Multi-Instance GPU)** . " " GPU instance .
- RAN/AI dynamic scenario .

6.2

- RAN workload **99%** — RAN .
- RAN **100%** — off-peak AI workload .
- Peak RAN SAC AI RAN → priority-weighted reward .
- AI workload RAN , .

, RAN .

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( ) , .

1. : RAN (L1/MAC) , MIG . dataset .
2. : task MIG → fractional/variable allocation (PRB/TB) .
3. **NRT-RIC** : Y1/E2 **10 ms~** . " " sub-TTI .
4. **AI workload** : AI task (token, attention) " MIG " .

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8 Gap

AI-RAN **AI-and-RAN(/)** . **importance-aware MAC scheduling AI-for-RAN( )** , **·granularity** .

8.1

	(CAORA)	
	NRT-RIC xApp + E2E orchestrator	GPU-collocated **L2 MAC scheduler**
	NRT-RIC (10 ms~), Y1/E2	**sub-TTI**,

	(CAORA)	
	MIG (GPU)	**PRB / TB** ( )
AI workload	(MIG)	**token importance**(entropy · attention) scheduler utility
"AI-and-RAN"	GPU RAN+AI **compute**	offloaded VLM RAN
		importance-aware
	,	**end-to-end task utility per resource**
	Python	Sionna → cuMAC on GH200, ROS2

8.2

- **AI workload** : AI workload *GPU* . use case multi-robot VLM offloading *uplink* . "AI RAN GPU " , "AI RAN " .
- : NRT-RIC/Y1 , sub-TTI externalize (GPU-collocated MAC) .
- : monitoring xApp , · **token** **proactive** **PRB** . .

8.3 Positioning

- predictive AI-and-RAN orchestration , **convergence backbone** , **macro** .
- (arXiv:2507.09124, *Proactive AI-and-RAN Workload Orchestration*) , reactive/ proactive gap anchoring .

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" RAN AI GPU(MIG) " NRT-RIC + SAC , AI-and-RAN . : RAN (VLM offloading) **token** **sub-TTI MAC** **importance-aware** — (micro) .